

HW2

Sam Olson

Q1: 1.3

(i)

Show that $\mathcal{F}_6 = \{A \in \Omega : A \text{ is countable or } A^c \text{ is countable}\}$ from example 1.1.4 is a σ -algebra.

(ii)

Define μ on $\mathcal{F}_6$ as $\mu(A) = 1$ if A is uncountable and $\mu(A) = 0$ if A is countable. Is μ a measure on $\mathcal{F}_6$? Justify.

Q2: 1.8

Let $\Omega \equiv \{1, 2, \dots\} = \mathbb{N}$ and $A_i \equiv \{j : j \in \mathbb{N}, j \geq i\}$, $i \in \mathbb{N}$. Show that $\sigma(\mathcal{A}) = \mathcal{P}(\Omega)$ where $\mathcal{A} = \{A_i : i \in \mathbb{N}\}$.

Q3: 1.10

Show that in Example 1.1.6, $\mathcal{O}_j \subset \sigma(\mathcal{O}_i)$ for all $1 \leq i, j \leq 4$.

- $\mathcal{O}_1 = \{(a_1, b_1) \times (a_2, b_2) : -\infty \leq a_1 < b_1 \leq \infty, -\infty \leq a_2 < b_2 \leq \infty\}$
- $\mathcal{O}_2 = \{(-\infty, x_1) \times (-\infty, x_2) : -\infty < x_1, x_2 < \infty\}$
- $\mathcal{O}_3 = \{(a_1, b_1) \times (a_2, b_2) : a_1 < b_1, a_2 < b_2, a_1, a_2, b_1, b_2 \in \mathbb{Q}\}$
- $\mathcal{O}_4 = \{(-\infty, x_1) \times (-\infty, x_2) : x_1, x_2 \in \mathbb{Q}\}$

(Okay to do for $k = 2$, i.e. $\mathbb{R}^2$.)

Q4: 1.18

Let $\Omega \equiv \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\Omega)$, and $A_n = \{j : j \in \mathbb{N}, j \geq n\}$, $n \in \mathbb{N}$. Let μ be the counting measure on $(\Omega, \mathcal{F})$. Verify that $\lim_{n \rightarrow \infty} \mu(A_n) \neq \mu(\cap_{n \geq 1} A_n)$.

Q5: 1.19

Let Ω be a nonempty set and let $\mathcal{C} \subset \mathcal{P}(\Omega)$ be a semialgebra. Let

$$\mathcal{A}(\mathcal{C}) \equiv \left\{ A : A = \bigcup_{i=1}^k B_i : B_i \in \mathcal{C}, i = 1, 2, \dots, k, k \in \mathbb{N} \right\}$$

(i)

Show that $\mathcal{A}(\mathcal{C})$ is the smallest algebra containing $\mathcal{C}$.

(ii)

Show also that $\sigma\langle\mathcal{C}\rangle = \sigma\langle\mathcal{A}(\mathcal{C})\rangle$.

Q6: 1.20

Let μ^* be as in (3.1) of Section 1.3:

Given a measure μ on a semialgebra $\mathcal{C} \subset \mathcal{P}(\Omega)$, the outer measure induced by μ is the function μ^* , defined on $\mathcal{P}(\Omega)$ as:

$$\mu^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \mu(A_n) : \{A_n\}_{n \geq 1} \subset \mathcal{C}, A \subset \bigcup_{n \geq 1} A_n \right\}$$

Verify:

(3.4)

$$\mu^*(\emptyset) = 0$$

(3.5)

$$A \subset B \Rightarrow \mu^*(A) \leq \mu^*(B)$$

(3.6)

For any $\{A_n\}_{n \geq 1} \subset \mathcal{P}(\Omega)$,

$$\mu^* \left(\bigcup_{n \geq 1} A_n \right) \leq \sum_{n=1}^{\infty} \mu^*(A_n).$$

(Hint: Fix $0 < \epsilon < \infty$. If $\mu^*(A_n) < \infty$ for all $n \in \mathbb{N}$, then find, for each n , a cover $\{A_{nj}\}_{j \geq 1} \subset \mathcal{C}$, $A_n \subset \bigcup_{j \geq 1} A_{nj}$ such that $\sum_{j=1}^{\infty} \mu(A_{nj}) \leq \mu^*(A_n) + \frac{\epsilon}{2^n}$.)

7. Chapter 1, Problem 1.24

Verify that $\mathcal{C}_2$, defined in (3.11), is a semialgebra.

Q8: Extra

Consider the following function:

$$F(x) = x \mathbb{I}\{0 < x < 1\} + \mathbb{I}\{x \geq 1\}, \quad x \in \mathbb{R}$$

(i)

Is this $F(\cdot)$ non-decreasing, right-continuous? Check.

(ii)

Define suitably a Lebesgue-Stieltjes measure μ_F on a suitable semialgebra $\mathcal{C}$. Do not have to prove $\mathcal{C}$ is a semi-algebra or μ_F is a measure.

(iii)

Prove that $\sigma\langle\mathcal{C}\rangle = \mathcal{B}(\mathbb{R})$. (Can use the fact that $\mathcal{B}(\mathbb{R}) = \sigma\langle\{\text{open sets}\}\rangle = \sigma\langle\mathcal{O}_1\rangle = \sigma\langle\mathcal{O}_2\rangle = \sigma\langle\mathcal{O}_3\rangle = \sigma\langle\mathcal{O}_4\rangle$ as in 1.10.)

(iv)

Argue that there exist a μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ that extends μ_F defined above. Do not need to prove this step - just state a result for your answer.

(v)

Is this extension unique? (i.e. if μ' is another extension of μ_F , then is it true that $\mu(A) = \mu'(A)$, for all $A \in \mathcal{B}(\mathbb{R})$?). Justify your answer (no proof needed). What is the “common name” for this measure μ ?